DO NOW — RECALL

Lesson 128 · Run-Length Encoding: Encoding and Decod

Name: _____ Class: _____ Date: _____

1. When must lossless compression be used?

2. In a queue, what does FIFO stand for?

3. What is a function parameter?

4. In a BST, which side holds a smaller value?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

When exact reconstruction is necessary, such as executable code.

2. Model answer

First in, first out.

3. Model answer

An input used by a subroutine.

4. Model answer

The left subtree.

By the end, you can…

• Group identical adjacent symbols, not identical symbols separated by others.

• Record every run as count and symbol consistently.

• Count storage using an explicit format, including counts and delimiters.

Vocabulary — know these terms

RLE — Encodes consecutive equal symbols as a symbol and count.

run — A consecutive sequence of identical symbols.

overhead — Extra data required by an encoding scheme.

lossless — Decoding reproduces the original exactly.

NEW LEARNING

Read, annotate and explain

Identify runs

Group identical adjacent symbols, not identical symbols separated by others.

Encode

Record every run as count and symbol consistently.

Check cost

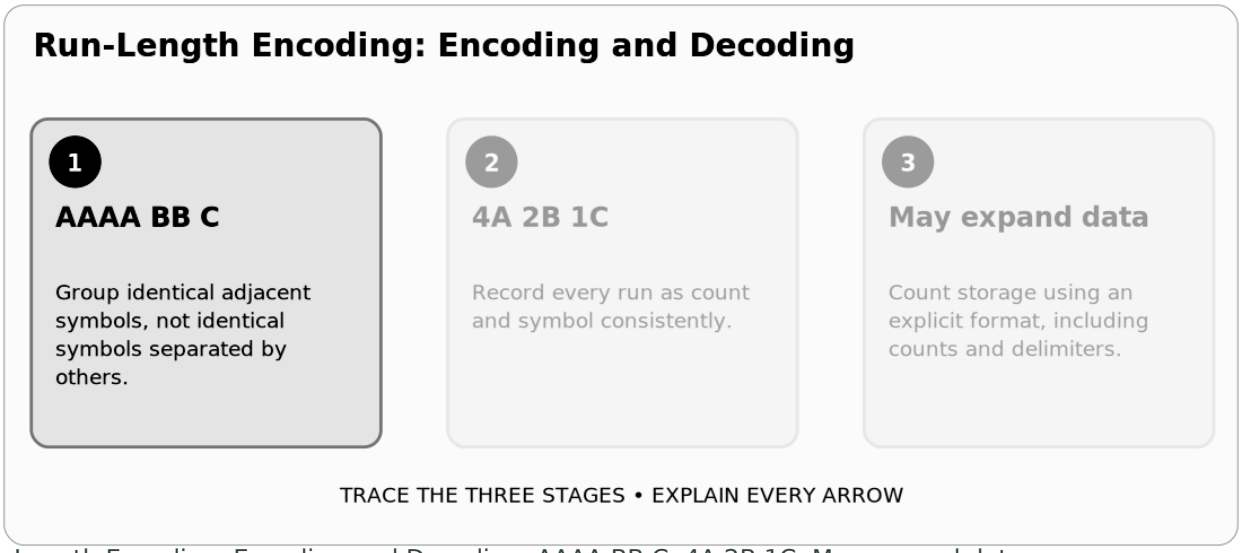
Count storage using an explicit format, including counts and delimiters.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

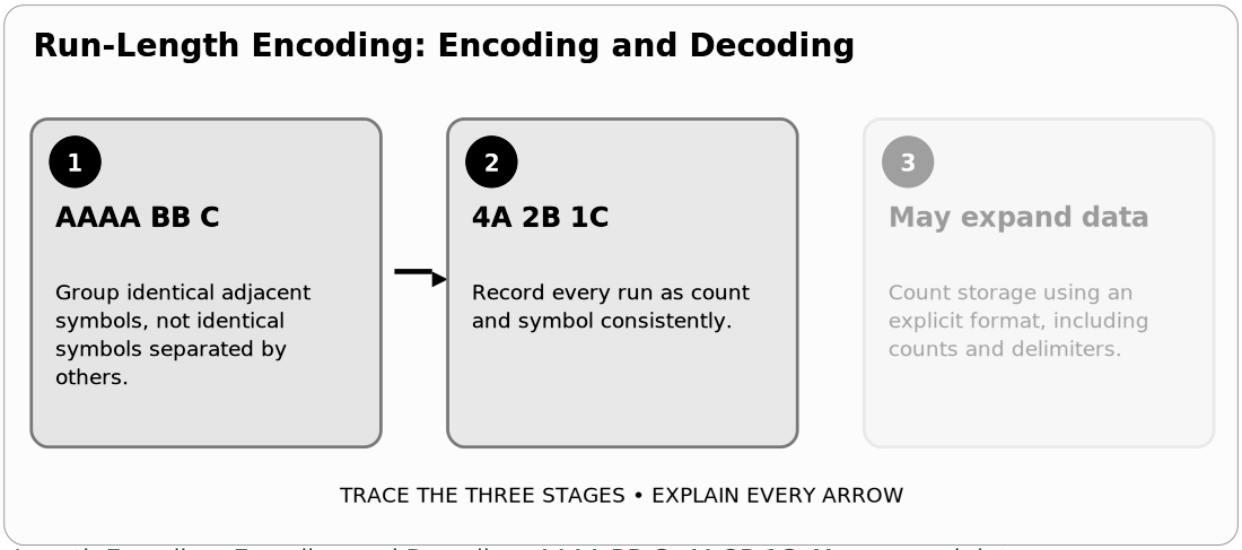
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Run-Length Encoding: Encoding and Decoding: AAAA BB C; 4A 2B 1C; May expand data

New Learning

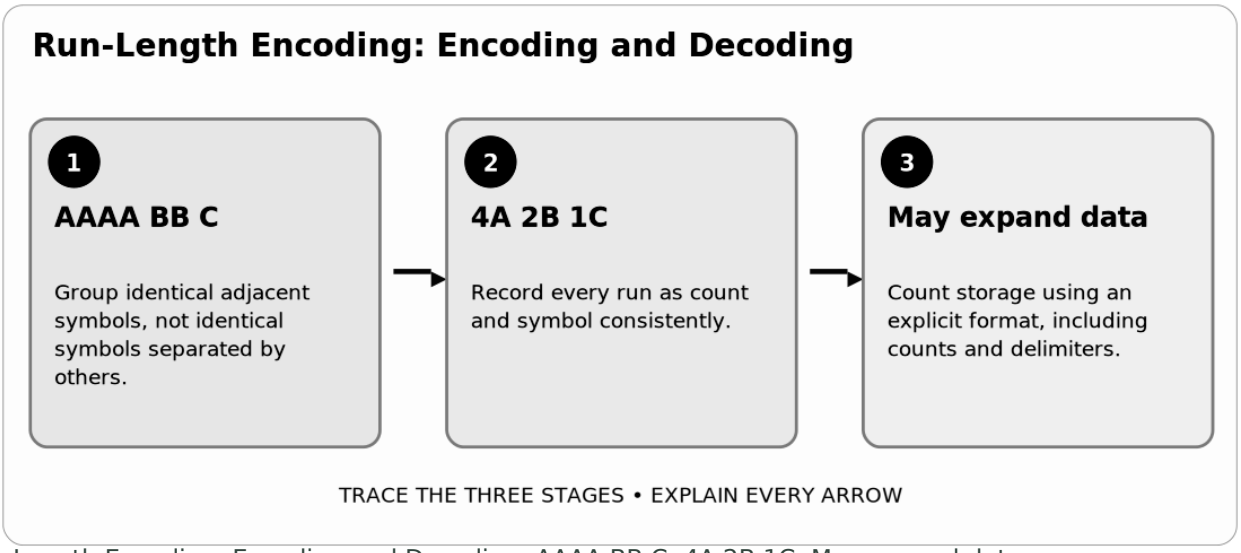


Run-Length Encoding: Encoding and Decoding: AAAA BB C; 4A 2B 1C; May expand data

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Run-Length Encoding: Encoding and Decoding: AAAA BB C; 4A 2B 1C; May expand data

I DO — WORKED EXAMPLE

Follow the worked method

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

AAAAA | BB | CCC

Predict the first step from the prompt.

(5,A)(2,B)(3,C)

Explain the intermediate result or decision.

Decode original

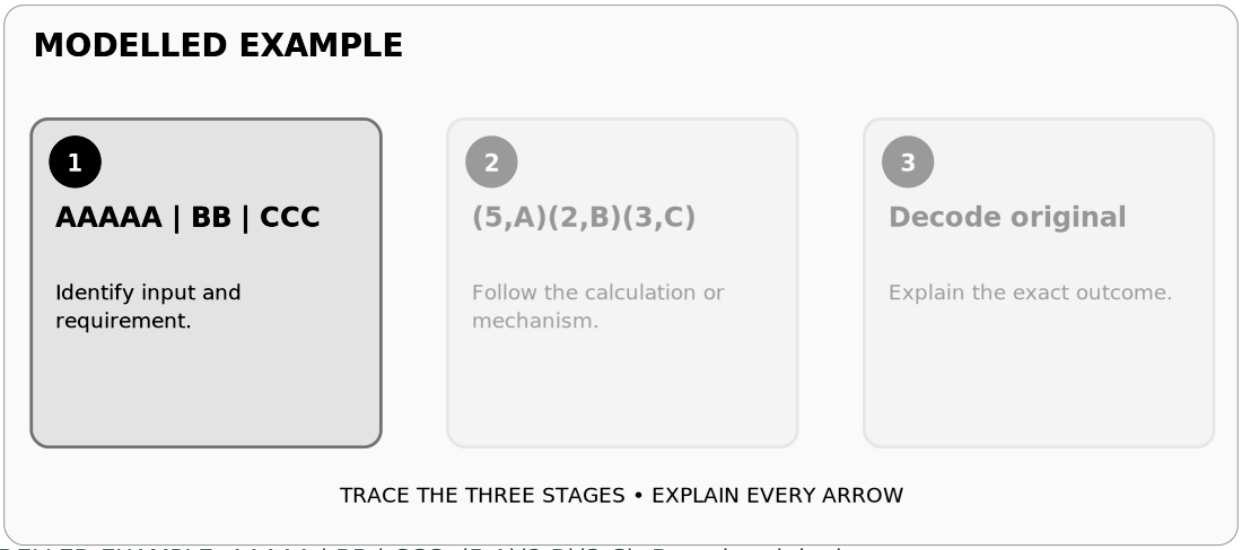
AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

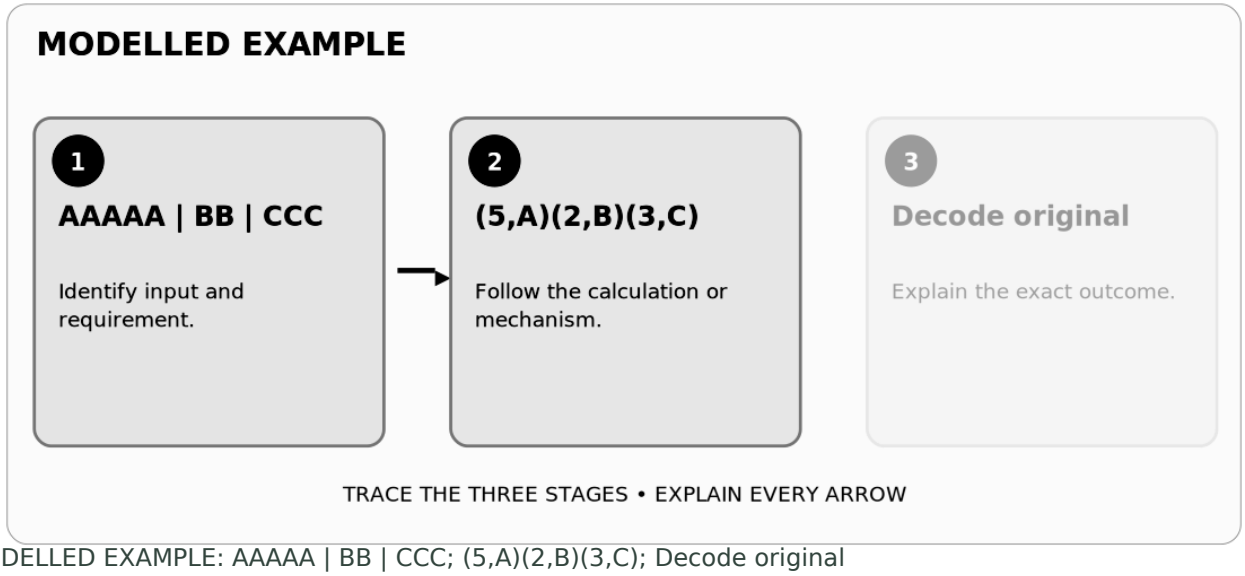
Predict the first step from the prompt.



MODELLED EXAMPLE: AAAAA | BB | CCC; (5,A)(2,B)(3,C); Decode original

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

Predict the first step from the prompt.



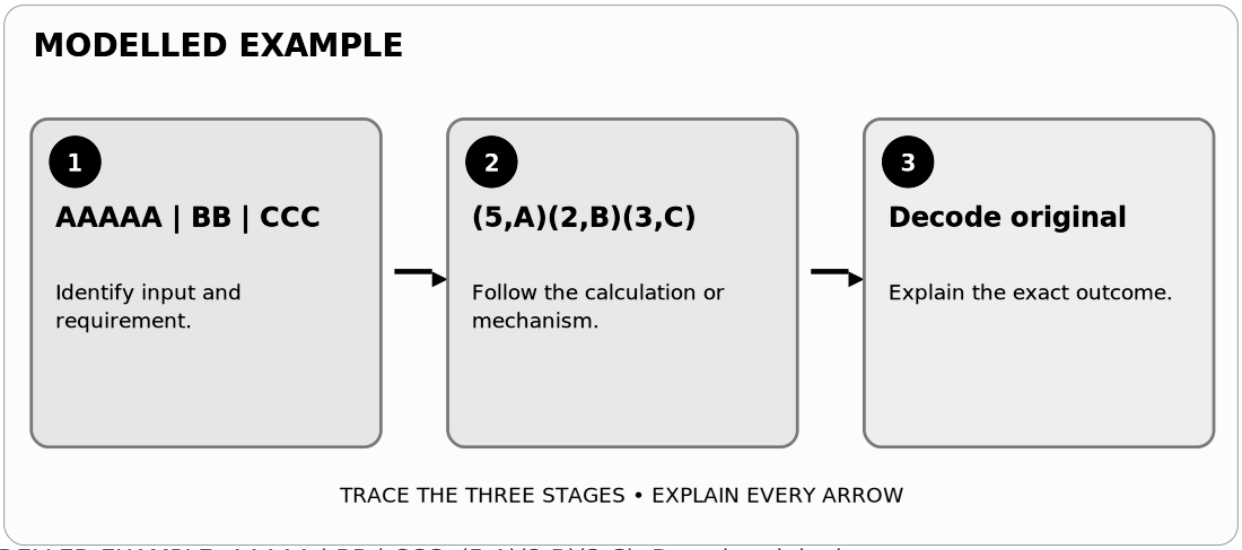
MODELLED EXAMPLE: AAAAA | BB | CCC; (5,A)(2,B)(3,C); Decode original

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

Predict the first step from the prompt.



MODELLED EXAMPLE: AAAAA | BB | CCC; (5,A)(2,B)(3,C); Decode original

WE DO — GUIDED PRACTICE

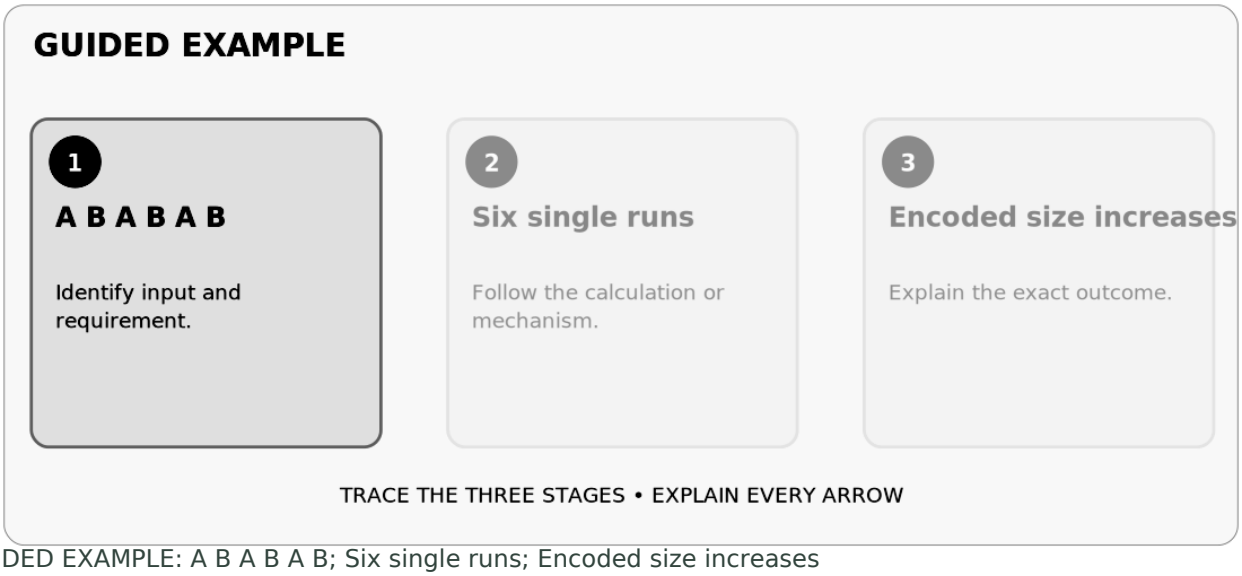
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Encode ABABAB as count-symbol pairs, and explain whether RLE helps.

Predict the first step from the prompt.



GUIDED EXAMPLE: A B A B A B; Six single runs; Encoded size increases

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: A B A B A B Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply RLE to AAAABBBCCAAAA using count-symbol pairs. [6]

Question 2

Decode (3,X)(1,Y)(2,X) and explain why runs must retain order. [5]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A bitmap row has 40 alternating pixels. Explain why RLE might increase file size and identify a better-suited pattern. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RLE always reduces the size of any input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

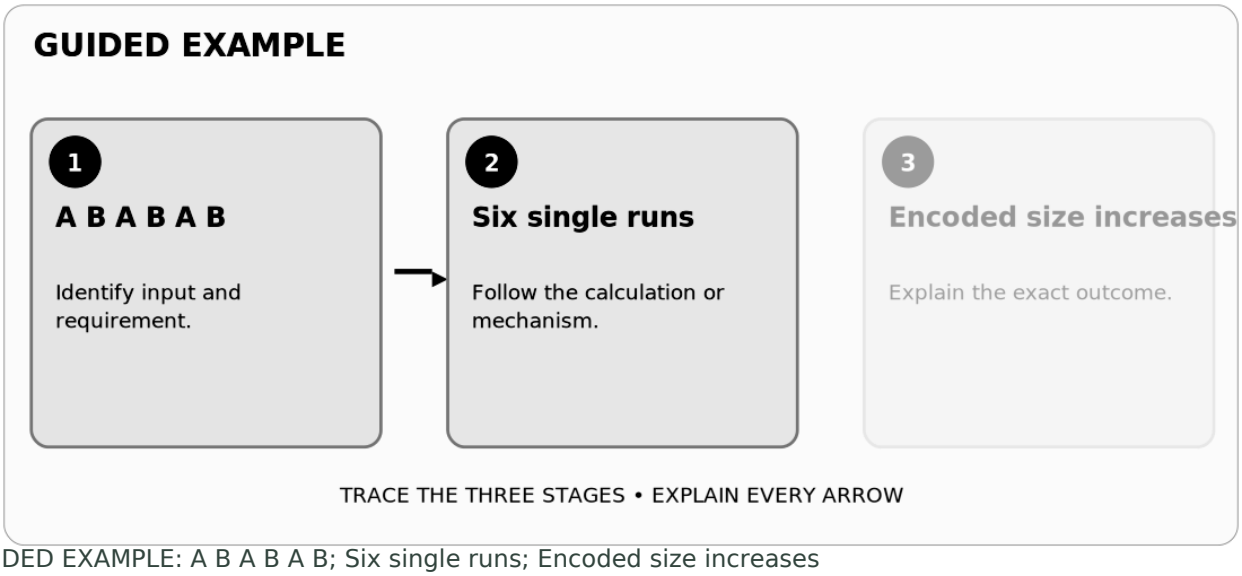
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Encode ABABAB as count-symbol pairs, and explain whether RLE helps.

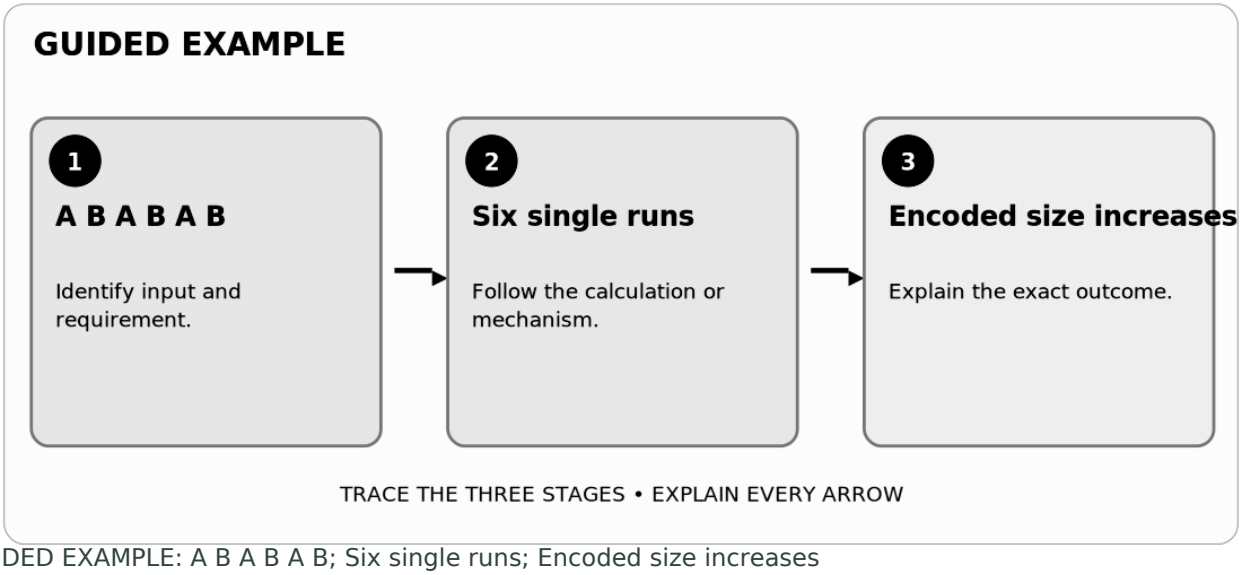
Predict the first step from the prompt.



GUIDED EXAMPLE: A B A B A B; Six single runs; Encoded size increases

Encode ABABAB as count-symbol pairs, and explain whether RLE helps.

Predict the first step from the prompt.



GUIDED EXAMPLE: A B A B A B; Six single runs; Encoded size increases

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Six single runs

Explain the intermediate result or decision.

Encoded size increases

Six runs of length 1 give (1,A)(1,B) repeated three times. A simple pair encoding uses 12 units for a 6-character input; RLE expands it, ignoring headers.

IMPROVE: Explain one correction to your own reasoning.